

CYBERVAST

PROFESSIONAL NETWORK SECURITY ASSESSMENT REPORT

Task 1A — Protocol Analysis & Traffic Capture

Week 1 Internship Milestone: Network Security Analyst

Prepared by: Ayomikun Olutoye

Cybersecurity Intern | Organisation: Cybervast

Report Version: 1.0 — FINAL

Table of Contents

Table of Contents..... 2

1. Executive Summary 3

2. Technical Environment & Laboratory Architecture..... 3

 2.1 Subnet Inventory & Node Configuration..... 3

 2.2 Architecture Rationale 4

3. Methodology & Packet Capture Execution 4

 3.1 Phase 1 Overview — Protocol Analysis & Traffic Capture 4

 3.2 Packet Capture Command Execution 4

4. Audit Findings: Protocol Analysis & Error Discoveries..... 5

 4.1 Wireshark Display Filter Syntax Error 5

 4.2 DNS Resolution Failures & ICMP Error Storm 6

 4.3 Critical Finding — Cleartext Administrative Credential Leakage 8

5. Risk Assessment Matrix10

6. Remediation Recommendations10

 6.1 Immediate — Eliminate Cleartext Administrative Sessions10

 6.2 Short-Term — Remediate DNS Infrastructure10

 6.3 Operational — Analyst Process Improvement11

7. Conclusion11

1. Executive Summary

This report documents the complete methodology, tooling, findings, and remediation recommendations arising from the Week 1 Protocol Analysis and Traffic Capture audit (Task 1A) conducted against Cybervast's private sandbox network segment. The engagement was executed by Ayomikun Olutoye in the capacity of Cybersecurity Intern.

At the outset of the assessment, the target environment lacked fundamental telemetry, network security baselines, and traffic enforcement controls. The primary objective was to transition this environment from an unmonitored, exposed configuration into a forensically visible and defensively hardened infrastructure — approached through a dual-perspective workflow that combined passive reconnaissance with active analytical engineering.

The Phase 1 audit (Task 1A) focused on passive frame analysis — evaluating how the system handled standard protocol suites under non-hardened conditions. Two critical findings emerged:

- DNS resolution failures caused by missing internal hostname mappings, generating cascading ICMP 'Port Unreachable' error storms.
- Cleartext credential leakage over an unencrypted FTP/Telnet administrative session, exposing privileged account credentials in full ASCII plaintext to any on-path observer.

⚠ CRITICAL FINDING — Credential Exposure

Administrative credentials were transmitted in cleartext over TCP Stream 0, fully readable via passive packet capture. This represents a CRITICAL risk requiring immediate remediation via TLS 1.3 enforcement.

These findings collectively confirm an unsafe baseline state and provide the evidentiary foundation for the subsequent host hardening, TLS 1.3 policy mandates, and automated SIEM containment structures implemented in later phases of this engineering sprint.

2. Technical Environment & Laboratory Architecture

To execute this security assessment without inducing operational risk or unauthorised data leakage, a fully isolated virtual laboratory engine was constructed using Oracle VirtualBox. The gateway architecture was managed via a virtualised pfSense security appliance, ensuring proper subnet segregation, DNS mapping, and centralised routing layout boundaries.

2.1 Subnet Inventory & Node Configuration

Component	Configuration Details
Network Gateway	pfSense Appliance — IP: 192.168.1.1 Handles DHCP allocations, DNS resolution, and baseline ACL enforcement

Analyst Workstation	Kali Linux — IP: 192.168.1.101 Deploys auditing, packet inspection, and administrative control operations
Target Infrastructure	Ubuntu 22.04 LTS — IP: 192.168.1.100 Hosts application environment, web services, and remote access daemons

2.2 Architecture Rationale

The sandboxed VirtualBox environment was chosen to ensure complete network isolation from production systems, while pfSense provided enterprise-grade gateway functionality including DHCP, DNS forwarding, and firewall rule enforcement. Kali Linux was selected as the analyst workstation due to its native support for industry-standard offensive and defensive security tooling, including Wireshark, tcpdump, and nmap.

3. Methodology & Packet Capture Execution

The assessment followed a structured, phased methodology designed to establish a reliable traffic baseline before moving into analytical fault-finding. All packet captures were performed using CLI-based utilities to preserve packet integrity and minimise host resource overhead, ensuring that captured data frames remained uncorrupted by GUI rendering processes.

3.1 Phase 1 Overview — Protocol Analysis & Traffic Capture

Phase	Details
Objective	Establish a definitive traffic baseline by capturing all raw network frames crossing the internal subnet under non-hardened conditions
Primary Tool	tcpdump (CLI packet capture) + Wireshark (deep packet inspection & stream reconstruction)
Capture File	AyomikunOlutoye Pcap File.pcap
Analyst Node	Kali Linux — 192.168.1.101
Target Node	Ubuntu 22.04 — 192.168.1.100
Protocols Analysed	DNS (UDP/TCP Port 53), ICMP, TCP (cleartext administrative sessions)

3.2 Packet Capture Command Execution

Packet capture was initiated natively on the Kali Linux analyst workstation using the tcpdump command-line utility. The -w flag was used to write raw frames directly to a structured .pcap file, ensuring complete frame fidelity without real-time rendering overhead:


```
# Packet capture command executed on analyst workstation (Kali Linux - 192.168.1.101)
sudo tcpdump -i eth0 -w AyomikunOlutoye_Pcap_File.pcap

# To capture all traffic across the internal subnet:
sudo tcpdump -i eth0 -nn -v -w AyomikunOlutoye_Pcap_File.pcap
```

The resulting .pcap file was subsequently loaded into Wireshark for deep packet inspection, protocol dissection, and stream reconstruction analysis.

4. Audit Findings: Protocol Analysis & Error Discoveries

4.1 Wireshark Display Filter Syntax Error

Upon loading the packet capture file into Wireshark and initiating protocol analysis, the first procedural issue encountered was a display filter syntax error triggered by an incorrect filter string entry.

Error Details

Field	Detail
Filter String Entered	dnsSS
Expected Filter	dns
Visual Indicator	Wireshark filter bar turned red; packet stream pane halted rendering
Root Cause	The string 'dnsSS' is not a recognised Wireshark BPF/display filter expression — Wireshark's filter parser is case-sensitive and protocol keywords must be lowercase
Resolution	The invalid string was cleared; correct lowercase filter 'dns' was applied, immediately restoring the packet stream view

The screenshot below captures this error state exactly as it appeared during the assessment — the red filter bar visible at the top of the Wireshark interface confirms the invalid filter rejection:

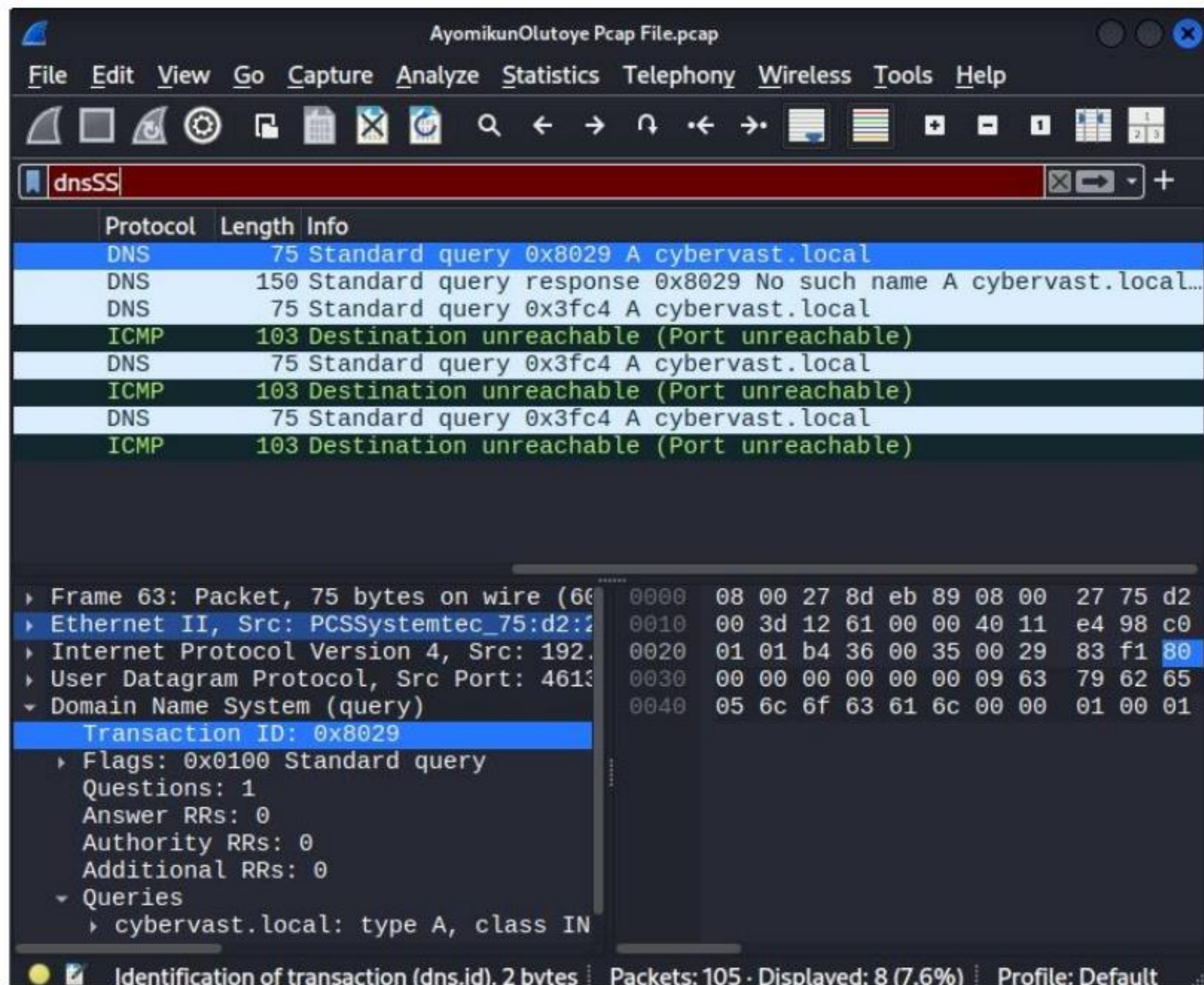


Figure 1: Wireshark displaying DNS/ICMP traffic with invalid 'dnsSS' filter string — filter bar shown in error state (red), before correction to lowercase 'dns'

i Filter Syntax Note

Wireshark display filters are fully case-sensitive. Protocol keywords such as 'dns', 'icmp', 'tcp', and 'http' must always be entered in lowercase. The BPF capture filter engine enforces the same rule. Incorrect casing causes a hard parse failure with no partial matching.

4.2 DNS Resolution Failures & ICMP Error Storm

Following resolution of the display filter error, correct application of the dns and icmp filters revealed a significant infrastructure fault: repeated DNS resolution failures for the internal domain cybervast.local, accompanied by a cascade of ICMP 'Destination Unreachable (Port Unreachable)' control messages.

DNS Packet Analysis — Frame 63

The packet stream confirmed the following sequence of events for Frame 63:

- Source: 192.168.1.101 (Analyst workstation) issued a standard DNS 'A' record query for cybervast.local
- Transaction ID: 0x8029 — Standard query with flags set to 0x0100
- DNS gateway response: 'No such name A cybervast.local' — confirming that the internal hostname was not registered in any zone file on the pfSense resolver

- Subsequent ICMP packets (Length: 103 bytes) were returned to source: 'Destination Unreachable (Port Unreachable)' — confirming that services were attempting to communicate with missing backend elements

Observation	Technical Detail
Affected Domain	cybervast.local
DNS Transaction ID	0x8029
DNS Query Flags	0x0100 — Standard query (recursion desired)
DNS Response	No such name — NXDOMAIN equivalent
ICMP Response Type	Destination Unreachable — Port Unreachable (Type 3, Code 3)
ICMP Packet Length	103 bytes
Root Cause	Internal DNS zone file for cybervast.local was missing or not propagated to the pfSense resolver; no A records were configured for internal hostnames
Impact	All services relying on hostname resolution failed; repeated retry loops generated unnecessary ICMP control traffic, creating a low-level error storm

The full packet capture view below shows the DNS query-response-ICMP error pattern as captured in Wireshark, with the packet detail pane confirming the transaction breakdown at Frame 63:

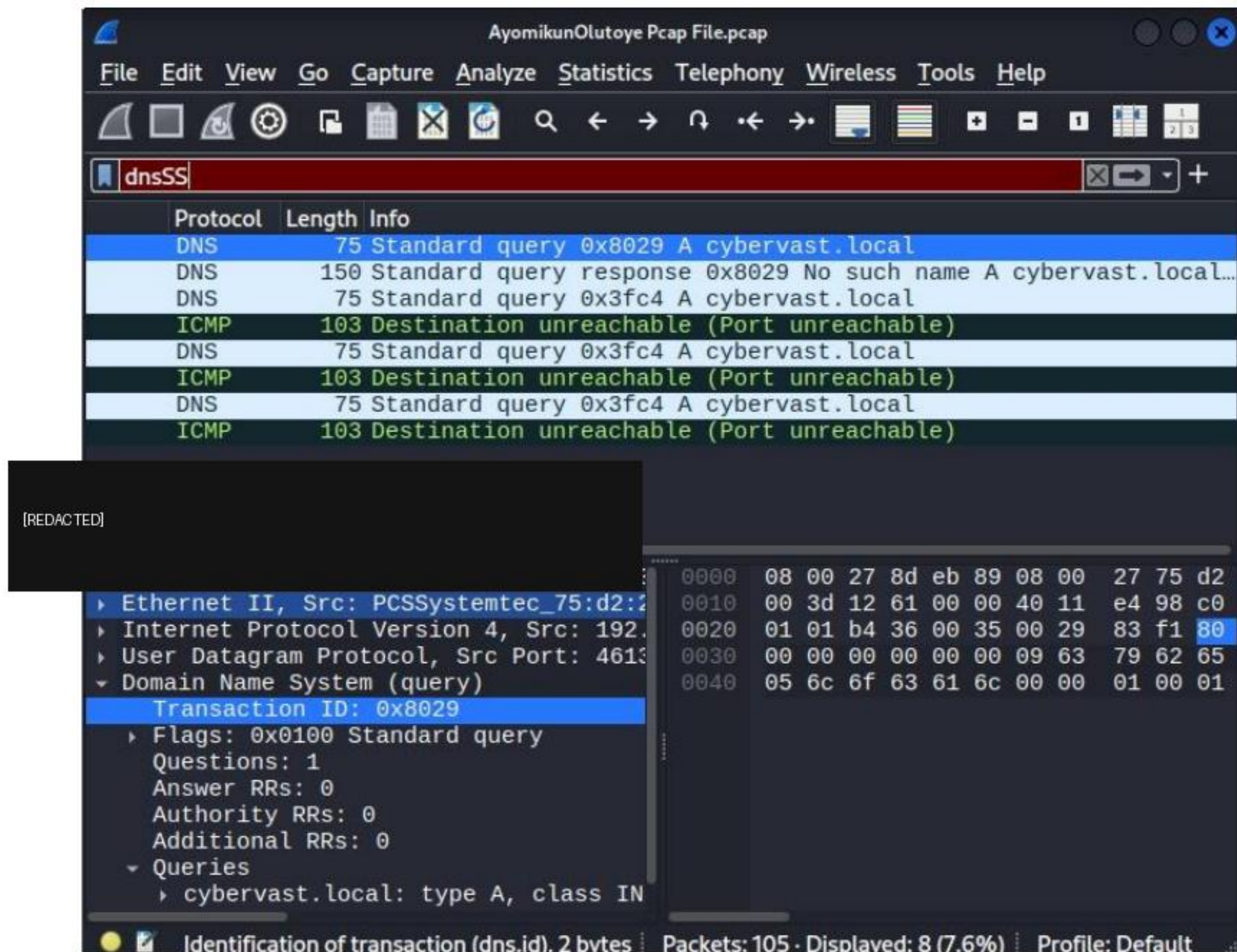


Figure 2: Wireshark packet capture — DNS resolution failures for cybervast.local (Transaction ID 0x8029) followed by ICMP Port Unreachable error responses. Corrected filter 'dns' applied.

⚠ Infrastructure Fault — DNS Zone Missing

No DNS A records exist for cybervast.local. This causes complete internal hostname resolution failure and generates repetitive ICMP error traffic. Remediation requires creating the forward lookup zone on the pfSense DNS resolver and adding appropriate host entries.

4.3 Critical Finding — Cleartext Administrative Credential Leakage

The most severe finding uncovered during this Phase 1 baseline audit was the discovery of plaintext administrative credentials being transmitted across the internal subnet over an unencrypted TCP session.

Using Wireshark's TCP stream reconstruction engine (Follow TCP Stream — tcp.stream eq 0), the very first raw TCP exchange captured in the .pcap file was reconstructed into human-readable ASCII format. Because the session relied on an unencrypted cleartext transport protocol (FTP or Telnet) rather than a cryptographically secure tunnel, every byte of upper-layer payload data was exposed in full ASCII plaintext.

Leaked Credential Data — TCP Stream 0

Field	Extracted Value
Wireshark Filter Used	Follow TCP Stream — tcp.stream eq 0
Capture File	[REDACTED]
Stream Direction	[REDACTED]
Conversation Size	46 bytes total
Leaked Username	USER intern_ayomikun
Leaked Password	PASS SecureAyomikun123!
Encoding	Standard ASCII — fully human-readable without decryption
Protocol	FTP or Telnet (cleartext administrative session over TCP)
Risk Rating	CRITICAL — Full credential compromise possible via passive interception

The forensic capture below documents the exact TCP stream reconstruction as displayed in Wireshark's Follow TCP Stream panel, showing the plaintext credentials in full:

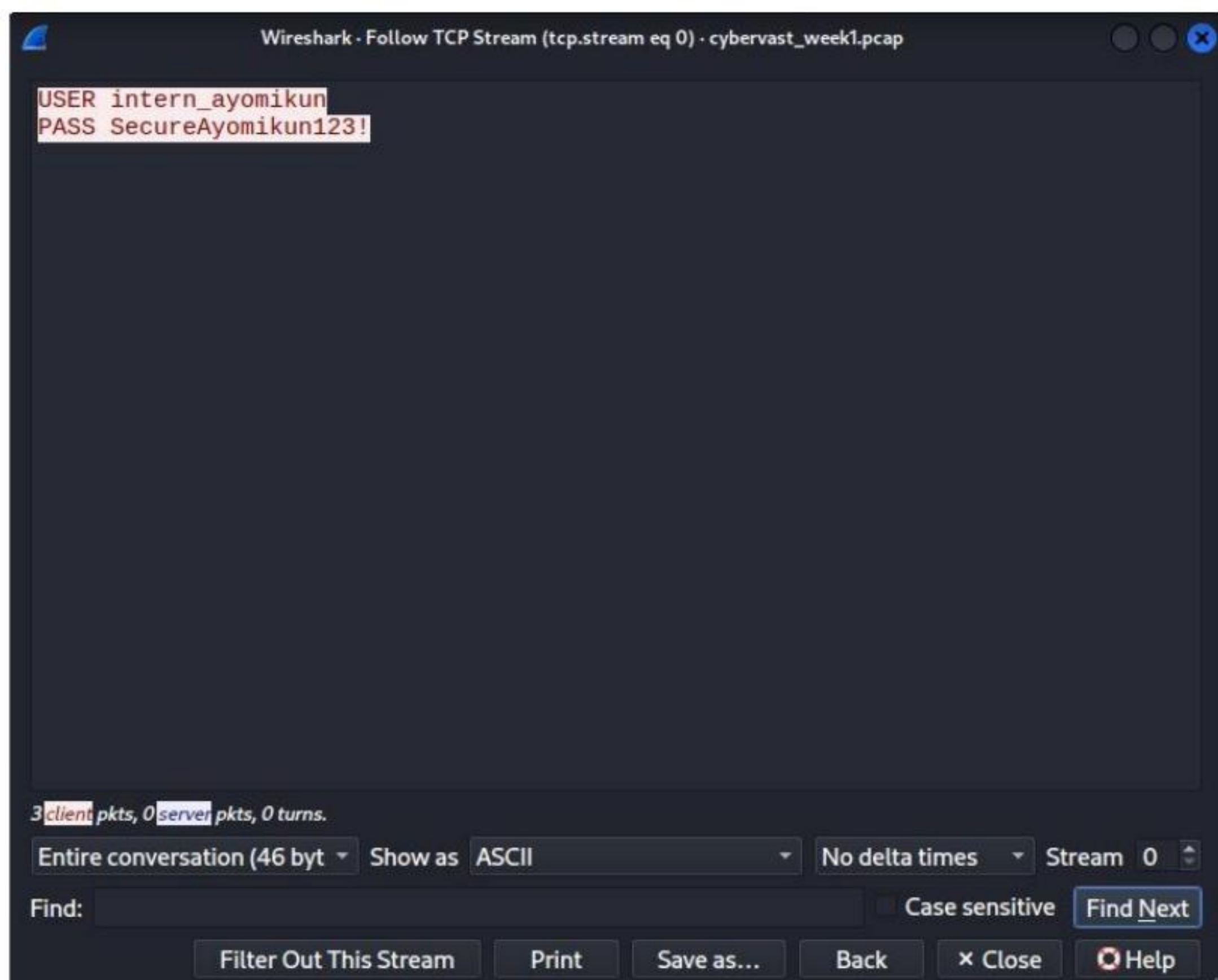


Figure 3: Wireshark — Follow TCP Stream (tcp.stream eq 0) — Cleartext credential leakage captured from cybervast_week1.pcap. Username 'intern_ayomikun' and password 'SecureAyomikun123!' visible in unencrypted ASCII plaintext.

● CRITICAL RISK — Credential Interception Confirmed

An on-path adversary positioned via passive network tap, ARP poisoning, or compromised adjacent node would have full visibility into these authentication packets with zero decryption overhead. Extraction of these credentials would allow complete account takeover and potential privilege escalation to root-level administrative access.

5. Risk Assessment Matrix

Finding	Risk Rating	Likelihood	Impact	Remediation Priority
Cleartext Credential Leakage (TCP Stream 0)	CRITICAL	High	High	Immediate
DNS Zone Missing — cybervast.local	HIGH	High	Medium	Short-term
ICMP Error Storm (Port Unreachable)	MEDIUM	Medium	Low	Short-term
Wireshark Filter Syntax Error	LOW (Operational)	Low	Low	Informational

6. Remediation Recommendations

6.1 Immediate — Eliminate Cleartext Administrative Sessions

- Disable all FTP and Telnet services on the target Ubuntu node immediately.
- Replace with encrypted alternatives: SSH (Secure Shell) for remote CLI access, SFTP or SCP for file transfer.
- Enforce TLS 1.3 across all web management interfaces and API endpoints.
- Implement network-level firewall rules on pfSense to block outbound traffic on TCP port 21 (FTP) and port 23 (Telnet).

6.2 Short-Term — Remediate DNS Infrastructure

- Create a forward lookup zone for cybervast.local on the pfSense DNS resolver (Unbound or dnsmasq).
- Add A records for all internal hostnames within the 192.168.1.0/24 subnet.
- Validate zone propagation using: `dig @192.168.1.1 cybervast.local A`
- Implement DNS query logging to detect future resolution failures early.

6.3 Operational — Analyst Process Improvement

- Standardise display filter templates: maintain a pre-validated Wireshark filter profile containing dns, icmp, tcp, http, and tls display filters to prevent syntax errors during live assessments.
 - Implement packet capture naming conventions and hash verification (MD5/SHA256) to preserve forensic chain-of-custody integrity for all .pcap files.
-

7. Conclusion

This Phase 1 protocol analysis and traffic capture assessment successfully documented the baseline security posture of Cybervast's private lab network segment prior to hardening. Two substantive vulnerabilities were identified and forensically evidenced:

- A critical cleartext credential leakage vulnerability caused by unencrypted administrative sessions traversing the internal subnet — confirmed via TCP stream reconstruction in Wireshark.
- A DNS infrastructure fault causing complete internal hostname resolution failure for cybervast.local, resulting in cascading ICMP error traffic.

Additionally, a procedural syntax error encountered during Wireshark filter configuration was documented, troubleshooted, and resolved — providing a training data point on correct display filter usage.

These findings collectively establish the documented 'before' baseline against which all subsequent hardening measures — TLS 1.3 enforcement, firewall rule implementation, DNS zone configuration, and SIEM integration — will be measured. The evidentiary pcap data and forensic captures included in this report will serve as the reference artefacts for Phase 2 comparative analysis.

Prepared by: Ayomikun Olutoye — Cybersecurity Intern, Cybervast

Report Version: 1.0 — FINAL | Task 1A — Protocol Analysis & Traffic Capture